Question 1 (Architecture vs. Organization)

Which term describes the operational units and their interconnections, encompassing the hardware details transparent to the programmer?

Answer Choices:

A. Computer Architecture

B. Computer Function

C. Computer Organization

D. Instruction Set Architecture (ISA)

Question 2 (Von Neumann Architecture)

The key characteristic of the **Von Neumann Architecture** is that it uses a single address space and bus for both:

Answer Choices:

A. Input/Output and Control Unit.

B. Arithmetic Logic Unit (ALU) and Registers.

C. Instructions (Program) and Data.

D. Cache Memory and Main Memory.

Question 3 (Parallelism vs. Concurrency)

Which term refers to the simultaneous execution of multiple instruction sequences on multiple processing units at the exact same physical time?

Answer Choices:

A. Concurrency

B. Synchronous evaluation

C. Threads

D. Parallelism

Question 4 (Flynn's Taxonomy)

A single processor that executes a single stream of instructions on a single stream of data belongs to which category in Flynn’s Taxonomy?

Answer Choices:

A. MISD

B. MIMD

C. SIMD

D. SISD

Question 5 (Memory Architecture Basics)

In memory architecture, which concept describes a system where all processing units share a single common memory space and access time is uniform for all processors?

Answer Choices:

A. NUMA

B. Distributed Memory

C. UMA

D. Hybrid Memory

Question 6 (Level of Parallelism)

What type of parallelism is achieved when multiple instructions are executed simultaneously within a single processor core, typically using techniques like pipelining and superscalar execution?

Answer Choices:

A. Data-Level Parallelism (DLP)

B. Task-Level Parallelism (TLP)

C. Bit-Level Parallelism (BLP)

D. Instruction-Level Parallelism (ILP)

Question 7 (Processor Component)

Which primary component of the Processor is responsible for performing arithmetic operations (like addition and subtraction) and logical operations (like AND, OR)?

Answer Choices:

A. Control Unit (CU)

B. Registers

C. Arithmetic Logic Unit (ALU)

D. Cache Memory

Question 8 (Multicore Programming)

In multicore programming, what is the technique of ensuring that work is distributed among available processing units so that no single core is overwhelmed while others are idle?

Answer Choices:

A. Concurrency

B. Synchronization

C. Load Balancing

D. Serial Code

Question 9 (GPU Basics)

GPGPU stands for:

Answer Choices:

A. General Processing Graphics Unit

B. General-Purpose Graphics Processing Unit

C. Globally Parallel Geometry Processor Unit

D. Graphics Program Generation Unit

Question 10 (HPC Scheduling)

In High-Performance Computing (HPC), which system component is responsible for managing the cluster resources (nodes, CPUs, memory) and efficiently allocating them to user jobs?

Answer Choices:

A. Interconnect Fabric

B. Cluster Architecture

C. Resource Manager / Job Scheduler

D. Compiler

Question 11 (Superscalar Architecture)

A processor is described as **Superscalar** if it can:

Answer Choices:

A. Execute instructions using a very long pipeline.

B. Execute multiple independent instructions in parallel in the same clock cycle using multiple execution units.

C. Switch quickly between multiple threads on a single core.

D. Process large vectors of data using a single instruction.

Question 12 (Instruction Set Architecture - ISA)

Which of the following attributes is defined by the **Instruction Set Architecture (ISA)**?

Answer Choices:

A. Cache line size and replacement policy.

B. The number of pipeline stages in the processor.

C. The number of available general-purpose registers and the instruction formats.

D. The cooling mechanism and packaging of the processor chip.

Question 13 (Interconnection Structures)

In a computer system, the primary function of the **Interconnection Structure** is to:

Answer Choices:

A. Decode instructions into micro-operations.

B. Provide the mechanism for the CPU, memory, and I/O modules to communicate.

C. Manage the flow of instructions through the processor pipeline.

D. Handle the physical heat dissipation from the processor.

Question 14 (Flynn's Taxonomy - Vector Processing)

A processor architecture designed for **Vector Processing**, where a single instruction operates on multiple data elements simultaneously (e.g., SIMD extension like AVX or NEON), fits primarily into which Flynn's Taxonomy category?

Answer Choices:

A. SISD

B. MISD

C. SIMD

D. MIMD

Question 15 (Memory Architecture - NUMA)

What is the primary trade-off/disadvantage introduced by a **NUMA (Non-Uniform Memory Access)** architecture compared to UMA?

Answer Choices:

A. All memory access requires a network interconnect, slowing down local access.

B. The system requires complex cache coherence protocols for all memory.

C. Access time to memory is slower if the data is located on a remote node's local memory.

D. It fundamentally restricts the system to having only two or four processors.

Question 16 (Parallel Programming - Dependency)

In parallel programming, a **Data Dependency** exists between two tasks if:

Answer Choices:

A. The two tasks are scheduled to run on the same processor core.

B. The output of one task is used as the input for the other task.

C. The tasks communicate via a shared network fabric.

D. The tasks are synchronized using a mutex or barrier.

Question 17 (Parallel Architecture)

An **Array Processor** is a specific type of parallel computer that best fits the **SIMD** model. Its primary structure consists of:

Answer Choices:

A. A single control unit and multiple independent processors, each with its own local memory, communicating via an interconnect.

B. Multiple independent processors, each with its own instruction and data stream (MIMD).

C. A single control unit directing a set of identical processing elements, all executing the same instruction simultaneously on different data.

D. A set of specialized hardware blocks designed for deep-learning acceleration.

Question 18 (Threads & Process)

In modern operating systems, multiple **Threads** within the same **Process** share which of the following resources?

Answer Choices:

A. Stack pointer and Program Counter.

B. Process ID (PID) and Address Space (Code, Data, and Heap).

C. Processor registers.

D. Thread-Local Storage (TLS).

Question 19

In the context of HPC job partitioning, which term refers to breaking down the data into smaller, independent chunks that can be processed simultaneously by different compute nodes?

Answer Choices:

A. Task-Level Parallelism

B. Data Decomposition

C. Instruction-Level Parallelism

D. Serial Code Isolation

Question 20

What is the primary role of the **SLURM** workload manager in an HPC cluster?

Answer Choices:

A. To manage the cluster's distributed file system.

B. To handle the networking traffic between nodes.

C. To allocate and manage compute resources for user jobs.

D. To compile parallel code using MPI and OpenMP.

Question 21

A modern NVIDIA GPU is best described as having a large number of simple, highly parallel processor cores, making it an excellent example of a high-end implementation of which Flynn's category?

Answer Choices:

A. SISD

B. MISD

C. SIMD / MIMD Hybrid

D. Pure MIMD

Question 22

The relationship between instruction count (IC), cycles per instruction (CPI), and clock cycle time (T_c) for calculating the total CPU execution time (T) is given by:

Answer Choices:

A. T = IC X CPI X Clock Frequency

B. T = IC / (CPI X T_c)

C. T = {C X CPI X T_c

D. T = T_c X (IC + CPI)

Question 23

Which of the following is the primary advantage of a hardware **Vector Processor** over a standard scalar processor using SIMD extensions?

Answer Choices:

A. It executes fewer instructions overall for a given task.

B. It eliminates the need for cache memory.

C. It allows for full utilization of ILP across all independent instructions.

D. It has built-in mechanisms to handle data streaming and memory access for large, contiguous data sets efficiently.

Question 24

A cluster system where each compute node has its own local memory (like a Distributed Memory system), but where the nodes are also capable of accessing a small, high-speed pool of shared memory through a specialized bus is best described as having a:

Answer Choices:

A. Pure UMA Architecture

B. Hybrid Memory Architecture

C. Non-Cached Architecture

D. Pure Distributed Memory Architecture

Question 25 (Parallel Processing Concepts)

Which term best describes the process of running multiple, completely unrelated applications simultaneously on a quad-core processor?

Answer Choices:

A. Data-Level Parallelism

B. Concurrency, primarily achieved through Task-Level Parallelism

C. Instruction-Level Parallelism

D. SIMD execution

Question 26 (Multicore Programming)

In multithreaded programming, what is the primary purpose of a **Synchronization Barrier**?

Answer Choices:

A. To block access to a critical section of code by using a lock.

B. To ensure that all threads reach a specific point in the code before any of them proceed further.

C. To force the operating system to perform a context switch.

D. To prevent the compiler from reordering instructions.

Question 27 (Level of Parallelism)

When a single instruction streams through a deep pipeline (e.g., Fetch, Decode, Execute, Memory, Write-back), this technique primarily exploits:

Answer Choices:

A. Bit-Level Parallelism

B. Data-Level Parallelism

C. Instruction-Level Parallelism

D. Task-Level Parallelism

Question 28 (Classification of Parallel Architectures)

Which architecture classification is characterized by having multiple control units, multiple processing units, and multiple instruction streams, making it the most flexible and complex?

Answer Choices:

A. SISD

B. SIMD

C. MISD

D. MIMD

Question 29 (Accelerators - FPGA)

In the context of specialized accelerators, what is the key advantage of using a **Field-Programmable Gate Array (FPGA)** over a standard CPU or GPU?

Answer Choices:

A. Higher clock frequency.

B. Ability to reconfigure the hardware logic itself to exactly match the computational task.

C. Significantly lower power consumption regardless of the workload.

D. Simpler and faster programming model using standard C++.

Question 30 (Symmetric Multiprocessor - SMP)

A key characteristic of a **Symmetric Multiprocessor (SMP)** system is that:

Answer Choices:

A. All processors are connected to the main memory using a crossbar switch.

B. All processors must be identical and have equal access to all peripheral devices and memory.

C. It utilizes a hierarchical bus structure with dedicated I/O processors.

D. It relies exclusively on distributed memory architecture (e.g., message passing).

Question 31

In a multiprocessor system with private caches, the **MESI Protocol** is used to enforce **Cache Coherence**. Which state in the MESI protocol indicates that a cache block is only available in the local cache and has been modified (is "dirty")?

Answer Choices:

A. Shared (S)

B. Valid (V)

C. Exclusive (E)

D. Modified (M)

Question 32 (Designing for Performance - Amdahl's Law)

A program takes 100 seconds to execute on a single core. The serial portion of the program is 10 seconds. According to Amdahl's Law, what is the maximum speedup achievable by running this program on an infinite number of cores?

Answer Choices:

A. 10 times

B. 90 times

C. 100 times

D. Infinite

Question 33 (GPU Architecture - CUDA/HIP)

In NVIDIA's CUDA architecture (or AMD's HIP), what is the term for a group of threads that execute the same instruction in lockstep on a Streaming Multiprocessor (SM)?

Answer Choices:

A. Thread Block

B. Grid

C. Kernel

D. Warp (or Wavefront in AMD)

Question 34 (Cache Coherence - Snooping)

The **Snooping Protocol** for cache coherence is typically used in which type of parallel architecture?

Answer Choices:

A. Distributed Memory Systems (e.g., HPC clusters with InfiniBand).

B. Tightly coupled Shared Memory Systems (e.g., SMP/UMA systems).

C. Architectures based on message passing only.

D. Purely SISD systems.

Question 35 (Parallel Programming - False Sharing)

In a multicore system using private caches, **False Sharing** is a performance-degrading phenomenon that occurs when:

Answer Choices:

A. Two independent variables are accessed by different cores but reside in the same cache line.

B. Two threads try to acquire the same mutex lock simultaneously.

C. A core attempts to read data that has not yet been written to main memory.

D. The compiler fails to detect all instruction-level parallelism.

Question 36 (Interconnects)

In an HPC cluster, a **Fat Tree** is a common topology for the interconnect fabric. A key characteristic of the Fat Tree is that:

Answer Choices:

A. Bandwidth decreases as one moves closer to the root (the core switches).

B. It provides uniform, non-blocking bandwidth between all compute nodes.

C. Every node is directly connected to every other node (full mesh).

D. It is only suitable for small, low-performance clusters.

Question 37 (Multicore Programming - Atomicity)

Which concept is used to ensure that a sequence of operations appears to an outside observer (other threads/cores) to occur instantaneously and indivisibly, preventing race conditions?

Answer Choices:

A. Synchronization Barrier

B. Instruction Reordering

C. Atomicity

D. Pipelining

Question 38 (GPU vs. CPU)

The architectural philosophy of a CPU compared to a GPU is often summarized as:

Answer Choices:

A. CPU: High Latency, High Throughput; GPU: Low Latency, Low Throughput.

B. CPU: Optimizing for Single-Thread Performance; GPU: Optimizing for Parallel Throughput.

C. CPU: Many simple cores; GPU: Few complex cores.

D. CPU: Mostly on-chip cache; GPU: Mostly on-chip registers.

Question 39 (Computer Components - I/O)

In a modern system, the main function of the **Direct Memory Access (DMA)** module is to:

Answer Choices:

A. Allow the CPU to directly access I/O ports without using the system bus.

B. Handle all interrupt requests from I/O devices.

C. Transfer data between I/O devices and main memory without continuous CPU intervention.

D. Store the boot firmware for the computer.

Question 40 (MISD Architecture)

The **MISD (Multiple Instruction, Single Data)** architecture is theoretically valid but rarely implemented in commercial, general-purpose computers. Which of the following best represents a practical, specialized use case resembling the MISD model?

Answer Choices:

A. A large-scale parallel database system.

B. A deep-learning training accelerator.

C. Pipelined cryptographic decryption systems where multiple functional units apply different steps to the same data stream.

D. A server running hundreds of independent virtual machines.

Coding/Conceptual Question 1 (Performance Calculation)

Task: Calculate the CPU execution time for a program with the following characteristics:

Instruction Count (IC) = 10^{10} instructions

Average Cycles Per Instruction (CPI) = 1.5

Clock Rate (Frequency) = 3 GHz

Answer:

Given:

$\text{IC} = 10^{10}$

$\text{CPI} = 1.5$

$\text{Frequency} = 3\text{ GHz} = 3 \times 10^9\text{ cycles/second}$

$\text{Clock Cycle Time } (T_c) = 1 / \text{Frequency} = 1 / (3 \times 10^9)\text{ seconds}$

Formula: $T = \text{IC} \times \text{CPI} \times T_c$

Calculation:

$$T = 10^{10} \times 1.5 \times \frac{1}{3 \times 10^9}$$

$$T = \frac{10^{10}}{10^9} \times \frac{1.5}{3}$$

$$T = 10 \times 0.5$$

$$T = 5\text{ seconds}$$

Final Answer: The total CPU execution time is **5 seconds**.

Justification:

The CPU execution time is derived from the fundamental CPU performance equation: $T = \text{IC} \times \text{CPI} \times T_c$. The clock cycle time ($T_c$) is the inverse of the clock frequency. Substituting the values yields a total execution time of 5 seconds.
